LI-FI PROJECT USING ARDUINO

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Introduction

Li-Fi (Light Fidelity) is a wireless communication technology that uses light to transmit data instead of radio waves.

It works by rapidly turning a light source ON and OFF, which represents digital data (1s and 0s).

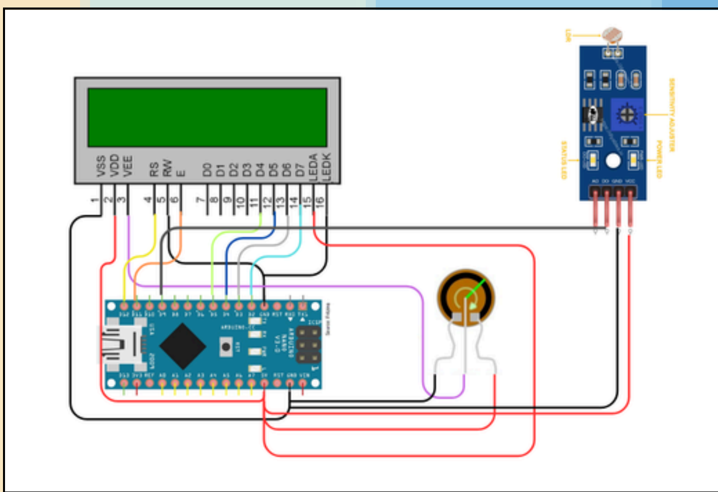
Objective

- To demonstrate **wireless communication** using visible light
- To transmit text messages using a smartphone flashlight
- To receive the signal using Arduino and LDR sensor
- To display the decoded message on an LCD display

Block Diagram

Li-Fi works on Visible Light Communication (VLC).

Circuit Diagram



Working

The mobile Li-Fi app sends a message using the phone flashlight as light signals. The LDR sensor detects these light changes and sends the signal to the Arduino. The Arduino converts the light pattern into binary data, decodes the message, and displays it on the LCD display.

Advantages

- Faster than Wi-Fi
- More secure
- Uses visible light spectrum
- No radio interference

Limitations

- Requires direct light path
- Cannot work through walls
- Affected by strong ambient light

Applications

- Underwater communication
- Smart homes
- Secure communication systems
- Hospitals (no RF interference)
- Aircraft communication
- Military communication